

Interview Prep: 60 Q&A

AI/ML Engineer | Data Scientist | Forward Deployed Engineer

Har sawal: Detail + Real-life Analogy + Interview mein kaise bolna hai

Prepared for Sumit | Interview: 12 October

Kaise padhein: Ratta mat maro. Pehle analogy samjho, phir detail, phir jawab apne words mein bolo. Jahan [] ya example diya hai, wahan sirf wahi bolo jo aapne sach mein kiya ho.

Part 1: AI/ML Engineer (20 Q)

1. Bias-variance tradeoff kya hai?

Detail: Bias matlab model ki galat assumptions se hone wali error. Model itna simple hai ki asli pattern pakad hi nahi paata (underfitting). Variance matlab model training data ke noise tak ko ratt leta hai, isliye naye data par fail hota hai (overfitting). Bias kam karoge to variance badhta hai, aur ulta bhi. Sweet spot beech mein hota hai.

Analogy: Do students. Student A ne sirf chapter 1 ka summary padha aur har sawal ka wahi jawab likhta hai (high bias). Student B ne pichle 5 saal ke paper ratt liye, naya sawal aate hi atak jaata hai (high variance). Sahi student concepts samajhta hai.

Interview mein bolo: Training error aur validation error compare karta hoon. Dono high hain to bias problem, training low aur validation high hai to variance problem.

2. Overfitting kaise rokte ho?

Detail: Overfitting tab hoti hai jab model training data par accha lekin naye data par kharab ho.

Tareeke: zyada data, cross-validation, regularization (L1/L2), dropout, early stopping, simple model aur feature selection.

Analogy: Cricket ka batsman jo sirf apne ghar ki pitch par practice karta hai. Wahan century maarta hai, lekin Australia ki bouncy pitch par fail. Solution: alag-alag pitch par practice (diverse data) aur har shot ratt ne ki jagah technique seekhna (regularization).

Interview mein bolo: Ek real example do: Mere model ka train accuracy 98% tha lekin validation 75%. Maine dropout aur early stopping lagaya aur gap kam hua.

3. Precision, Recall aur F1 kab use karte ho?

Detail: **Precision** = jitne ko positive bola, usme se kitne sach mein positive the. **Recall** = jitne sach mein positive the, usme se kitne pakde. **F1** = dono ka balanced score.

Analogy: Airport security. Har kisi ko roko (high recall, low precision) to koi chor nahi bachega lekin hazaron innocent pareshan honge. Sirf pakka shak hone par roko (high precision, low recall) to innocent bachenge lekin kuch chor nikal jaayenge. Bomb detection mein recall chahiye, spam filter mein precision.

Interview mein bolo: Metric business cost se decide hota hai. False negative costly hai to recall, false positive costly hai to precision.

4. Imbalanced data kaise handle karte ho?

Detail: Maan lo 1000 transactions mein sirf 10 fraud hain. Model sabko not fraud bol de to 99% accuracy, lekin kaam ka nahi. Solutions: SMOTE, undersampling, class weights, threshold tuning, aur accuracy ki jagah PR-AUC/F1/recall.

Analogy: Commentator jo 99 balls dot ball bolta hai aur 1 wicket miss kar deta hai. Accuracy 99% hai lekin asli important event miss ho gaya.

Interview mein bolo: Pehle metric badalta hoon, phir class weights, aur SMOTE sirf training set par lagata hoon, test set par kabhi nahi.

5. Gradient Descent aur Adam optimizer kya hain?

Detail: Gradient descent loss function minimize karne ka tareeka hai. Har step mein weights ko us direction mein update karte hain jahan loss sabse tezi se kam hota hai. Learning rate step ka size tay karta hai. Adam mein momentum aur adaptive learning rate (har parameter ka alag step size) dono hote hain.

Analogy: Kohre mein pahaad se ghati mein utarna. Aapko sirf pair ke neeche ki dhalaan dikhti hai. Learning rate bada hua to kood kar paar nikal jaaoge, chhota hua to subah tak pahunchoge. Adam ek smart watch jaisa hai: slope steep ho to dheere, flat ho to tez.

6. Transformer aur self-attention samjhao.

Detail: Self-attention har word ko sentence ke baaki words se compare karke decide karta hai ki kis par kitna dhyan dena hai. Har token ke liye Query (main kya dhundh raha hoon), Key (mere paas kya hai) aur Value (actual info) banta hai. Query-Key match score batata hai ki kis Value ko kitna weight milega. RNN ki tarah sequential nahi, poora sentence parallel process hota hai.

Analogy: Sentence: Rahul ne Amit ko kitab di kyunki wo pareshan tha. Wo kaun? Insaan ka dimaag pichle words dekhta hai. Self-attention bhi wo ke liye baaki saare words ko score deta hai aur sahi context utha leta hai.

7. RAG kya hai aur fine-tuning se kab behtar hai?

Detail: RAG (Retrieval-Augmented Generation) mein user ke sawal ke time relevant documents dhundhe jaate hain aur LLM ko context mein diye jaate hain. Fine-tuning mein model ke weights change hote hain.

RAG: knowledge baar-baar badalti ho, private data ho, citations chahiye.

Fine-tuning: tone/style/format badalna ho ya specialized behavior seekhna ho.

Analogy: RAG = open-book exam (kitab se sahi page nikal kar jawab). Fine-tuning = mahino coaching taaki soch ka tareeka hi badal jaaye. Syllabus har hafte badalta hai to open-book better hai.

8. RAG ki quality kaise improve karoge?

Detail: Zyada tar problem retrieval mein hoti hai. Improvements:

- **Chunking:** sahi size aur overlap, ya semantic chunking.
- **Hybrid search:** keyword (BM25) + embeddings.
- **Reranker:** top 20 nikal ke better model se top 5 chunno.
- **Query rewriting** aur **metadata filters**.
- **Eval:** retrieval ko alag naapo (recall@k).

Analogy: Library mein librarian ko kitab dhundhne bhejo. Kitaab ke tukde galat kaat diye (galat chunking) to context toot jaayega. Sirf title se dhundhe (sirf keyword) to meaning miss ho sakta hai. Isliye keyword aur meaning dono se dhundhna sikhao, phir senior shortlist check kare (reranker).

9. Hallucination kaise kam karoge?

Detail: Hallucination matlab LLM ka confident lekin galat jawab dena. Reduce karne ke tareeke: RAG se grounding, prompt mein agar context mein nahi hai to bolo pata nahi, citations maangna, low temperature, structured output + validation, aur eval set par regular testing.

Analogy: Over-confident dost jo kabhi pata nahi nahi bolta. Rasta batayega, galat ho tab bhi. Solution: usse bolo sirf wahi bolo jo is map mein likha hai aur batao map ke kis hisse se.

10. LoRA aur QLoRA kya hain?

Detail: Bade model ko poora fine-tune karna bahut mehenga hai. LoRA mein original weights freeze rehte hain aur sirf chhote low-rank adapter matrices train hote hain. Memory aur cost kam. QLoRA mein base model 4-bit quantized hota hai aur uske upar LoRA lagta hai, to ek single GPU par bhi fine-tune ho jaata hai.

Analogy: 1000-page ki kitab mein ek chhoti baat badalni hai. Poori dbara likhne ki jagah ek sticky note chipka do (LoRA adapter). QLoRA mein kitab ko pehle compress kiya, phir sticky note lagaya.

11. LLM application ko evaluate kaise karoge?

Detail: 1. Golden test set banao (sawal + expected jawab, 50-200).

2. Retrieval metrics (recall@k) aur answer metrics (correctness, faithfulness, relevance) alag naapo.

3. LLM-as-judge use karo, human spot-check ke saath.

4. Production mein user feedback aur logging.

5. Har change ke baad regression test.

Analogy: Chef jo har baar mujhe to swaad aaya bolke menu badalta hai, ye evaluation nahi. Sahi tareeka: fixed tasting panel, scoring sheet, aur har naye recipe ko usi panel se score karwana.

12. Embeddings aur Vector DB kaise kaam karte hain?

Detail: Embedding text ko numbers ke vector mein badalta hai jahan similar meaning wale texts paas-paas hote hain. Vector DB in vectors ko store karta hai aur query ke sabse nearest vectors dhundhta hai (cosine similarity). Speed ke liye ANN indexes (jaise HNSW) use hote hain.

Analogy: Bade mall ka map jahan similar dukaanein paas-paas hain. Sasta running shoe poochoge to seedha shoe section jaoge. Embedding wo map hai, vector DB wo guide hai jo nearest dukaan batata hai.

13. AI Agent kya hai aur kaun si problems aati hain?

Detail: Agent aisa LLM hai jo khud plan banata hai, tools call karta hai, result dekhta hai aur agla step decide karta hai (plan, act, observe). Problems: infinite loops, galat tool call, high cost/latency, errors ka accumulate hona. Solutions: max step limit, structured outputs, guardrails, logging, risky actions par human approval.

Analogy: Naya intern jise bola flight book karo. Khud website kholega, price compare karega, book karega. Guide na ho to galat date par book kar sakta hai ya 50 baar same page refresh karega. Isiye rules, limits aur payment se pehle approval.

14. Prompt Injection kya hai aur kaise bachoge?

Detail: Attacker document, email ya webpage mein hidden instruction daal deta hai (pichli instructions bhool jao aur confidential data bhejo). LLM use genuine instruction samajh sakta hai. Bachav: untrusted content ko instructions se alag treat karna, least-privilege tool access, output filtering, sensitive actions par human approval.

Analogy: Secretary boss ki dak kholti hai. Ek letter mein likha hai ye boss ka order hai, cheque book is address par bhej do. Achhi secretary samajhti hai ki ye letter ka content hai, boss ka order nahi.

15. Model ko production mein kaise deploy karoge?

Detail: Model ko API service mein wrap karo (FastAPI), Docker container banao, Kubernetes ya managed service (SageMaker/Vertex AI) par chalo. Saath mein autoscaling, health checks, logging, monitoring aur rollback plan. Real-time nahi chahiye to batch job (Airflow).

Analogy: Ghar par banayi recipe (notebook) ab restaurant mein bechni hai: standard packaging (Docker), counter par order (API), rush mein zyada chef (autoscaling), kitchen mein camera (monitoring), kuch kharab ho to purana menu wapas (rollback).

16. LLM app ka latency aur cost kaise kam karoge?

Detail: Caching, model routing (aasaan sawal chhote/sasta model ko), prompt chhota karna, streaming, batching, quantization aur max tokens limit.

Analogy: Courier company ke paas har parcel ke liye Boeing nahi hota. Local parcel bike se, city ke beech truck se, sirf urgent international jahaz se. Aasaan sawal chhote model se, mushkil bade model

se.

17. Data drift aur concept drift mein fark?

Detail: Data drift: input data ka distribution badal gaya (customers ki average age badal gayi).

Concept drift: input aur output ka relationship badal gaya (COVID ke baad log kya khareedte hain wo pattern hi badal gaya). Detection: PSI, KS test, live performance monitoring. Fix: retraining.

Analogy: Delhi ki sardiyon ke liye jacket sales predict karna seekha. Data drift = ab aap Chennai mein ho, customers alag. Concept drift = Delhi mein hi, lekin fashion badal gaya aur log jacket ki jagah hoodie lete hain.

18. MLOps pipeline kaisi hoti hai?

Detail: Data ingestion -> data validation -> training -> experiment tracking (MLflow) -> model registry -> CI/CD deploy -> monitoring -> drift hone par retraining trigger. Har step automated aur reproducible hona chahiye.

Analogy: Factory assembly line. Kachcha maal (data), quality check (validation), product banta hai (training), label lagta hai (versioning), shop tak jaata hai (deploy), customer complaints par design improve hota hai (monitoring aur retraining).

19. Quantization kya hai?

Detail: Model ke weights ko FP32/FP16 se INT8/INT4 jaise kam precision mein convert karna. Model chhota hota hai, memory kam lagti hai, inference tez hota hai, lekin accuracy mein thodi kami aa sakti hai.

Analogy: HD photo ko compress karke WhatsApp par bhejna. File chhoti, jaldi gayi, dekhne mein almost same; bahut zoom karoge to thodi quality kam dikhegi.

20. System design: company documents par chatbot kaise banaoge?

Detail: 1. **Ingestion:** documents parse, chunk, metadata add.

2. **Embedding + Vector DB** mein store.

3. **Retrieval:** hybrid search + reranker.

4. **LLM** ko context ke saath prompt.

5. **Guardrails:** hallucination check, PII filter, citations.

6. **Access control:** employee sirf wahi documents dekhe jiski permission hai.

7. **Eval, caching, logging aur feedback loop.**

Analogy: Smart company librarian. Naye documents sahi shelf par rakhta hai (ingestion), sawal par sahi kitaab dhundhta hai (retrieval), jawab ke saath page number deta hai (citations), aur check karta hai ki aap confidential almirah dekhne ke haqdaar ho ya nahi (access control).

Interview mein bolo: System design mein pehle clarifying sawal poocho: kitne documents, kitne users, latency, data kitna sensitive.

Part 2: Data Scientist (20 Q)

21. Central Limit Theorem (CLT) kya hai?

Detail: Kisi bhi distribution se baar-baar random samples lo (size 30+) aur har sample ka mean nikalo, to sample means ka distribution normal (bell curve) ke close ho jaata hai, chahe original data kaisa bhi ho. Isi wajah se confidence intervals aur hypothesis tests kaam karte hain.

Analogy: Chai ki tapri par har customer ka bill alag hai (Rs 10 se Rs 500). Roz ke 50 customers ka average bill likho. Kuch dino baad ye daily averages bell curve banate hain.

Interview mein bolo: CLT ki wajah se main population ke distribution ki chinta kiye bina sample mean par test laga sakta hoon.

22. p-value kya hota hai?

Detail: Agar null hypothesis sach hota, to itna extreme ya usse zyada extreme result milne ki probability. $p = 0.03$ ka matlab: agar koi asli fark nahi hota to ye result sirf 3% chance se aata. Ye galat hai ki hypothesis ke sach hone ki probability 97% hai. Ye sabse common galti hai.

Analogy: Dost kehta hai mera sikka fair hai. 10 baar uchalte ho, 9 baar heads. p-value poochta hai: sikka fair hota to 9 ya zyada heads ki kitni chance thi? Bahut kam (~1%), to shak hota hai, lekin ye proof nahi hai.

Interview mein bolo: p-value sirf evidence ki strength batata hai, effect ka size nahi. Isliye effect size aur confidence interval bhi dekhta hoon.

23. Type I aur Type II error kya hain?

Detail: Type I (false positive): asli mein fark nahi tha, humne bol diya fark hai. Alpha (0.05) iski probability control karta hai.

Type II (false negative): asli mein fark tha, humne pakda nahi. Power = 1 - beta.

Analogy: Court case. Type I = nirdosh ko saza de di. Type II = asli gunehgaar ko chhod diya.

Interview mein bolo: Medical test mein Type II zyada khatarnak hai (bimari miss), isliye wahan recall/power badhate hain.

24. A/B test kaise design karoge?

Detail: 1. Hypothesis aur primary metric tay karo.

2. Sample size calculate karo (power analysis).

3. Random assignment.

4. Fixed duration (kam se kam 1-2 poore weeks).

5. Test khatam hone ke baad analysis.

Galtiyan: peeking (beech mein result dekh ke rok dena), bahut saare metrics (multiple testing), novelty effect.

Analogy: Do chai wale ki recipe test karni hai. Dono randomly alag customers ko do, kafi customers tak chalao, beech mein A jeet raha hai bolke rok mat do.

25. Correlation aur causation mein fark?

Detail: Correlation sirf do cheezon ka saath-saath badalna hai. Causation matlab ek cheez doosri ko cause karti hai. Causation prove karne ke liye randomized experiment ya causal methods (difference-in-differences, propensity score matching, instrumental variables) chahiye.

Analogy: Garmi mein ice-cream sales aur doobne ke cases dono badhte hain. Ice-cream se koi doobta nahi. Dono ka asli reason garmi hai (confounder).

26. Missing values kaise handle karte ho?

Detail: Pehle samjho kyun missing hai: random (MCAR), kisi aur column par depend (MAR), ya us value ki wajah se (MNAR, jaise high income wale income batate hi nahi). Phir options: drop (agar bahut kam), mean/median/mode imputation, model-based imputation (KNN, iterative), ya is_missing indicator column.

Analogy: Class register mein kuch bachchon ki attendance khali hai. Pehle poocho: machine kharab thi, ya sirf Monday ko khali hai, ya jo bunk karte hain unhone khud nahi bhari? Reason ke hisaab se fix alag.

27. Outliers kaise handle karoge?

Detail: Detect: IQR, z-score, boxplot. Phir sabse zaroori sawal: error hai ya genuine? Error ho to fix/remove. Genuine ho (fraud ya VIP customer) to remove mat karo. Options: capping (winsorization), log transform, robust models (tree-based).

Analogy: Class ki average height mein ek entry 17 feet hai: typo, theek karo. Basketball team ka student 6.8 feet hai: genuine, hatana galat.

28. Data leakage kya hai?

Detail: Training mein aisi information aa jaana jo prediction ke time real life mein available nahi hogi: future ka data, target se bani feature, ya split se pehle scaling/imputation. Result: lab mein 99%, production mein flop. Bachav: time-based split, preprocessing sirf train par fit karna, har feature par poochna kya ye prediction ke time pata hogi?

Analogy: Exam se pehle paper leak ho gaya aur aapne jawab ratt liye. Result 100%, lekin naye sawal aaye to zero.

29. L1 aur L2 regularization mein fark?

Detail: Dono complexity par penalty lagate hain.

L1 (Lasso): weights ka absolute value penalize. Kuch weights exactly zero, automatic feature selection.

L2 (Ridge): weights ke square ko penalize. Weights chhote lekin zero nahi.

Analogy: Team banate ho. L1 = kuch players ko team se bahar kar dena. L2 = sabko rakhna lekin kisi ek ko bahut zyada power mat do.

30. Random Forest aur Gradient Boosting mein fark?

Detail: Random Forest (bagging): bahut saare trees independently, parallel mein; final answer voting/average. Variance kam.

Gradient Boosting (XGBoost, LightGBM): trees ek ke baad ek, har naya tree pichle ki galtiyan sudharta hai. Bias kam. Aksar zyada accurate, lekin tune karna padta hai aur overfit ho sakta hai.

Analogy: Random Forest = 100 doston se poochna aur majority ki baat maanna. Boosting = ek student jo har mock test mein apni pichli galtiyan dekh kar sudharta hai.

31. Cross-validation kyun, aur time series mein kya alag?

Detail: Ek hi train/test split lucky ya unlucky ho sakta hai. K-fold CV mein data K hisson mein baanta jaata hai, har baar ek hissa test, baaki train, aur average liya jaata hai. Time series mein random split nahi (future data past mein aayega, leakage). Wahan time-based split (expanding ya rolling window).

Analogy: Student ko ek mock test se judge mat karo, 5 ka average lo. Lekin share price predict karna ho to hamesha past se future predict karo.

32. ROC-AUC aur PR-AUC mein fark?

Detail: ROC-AUC true positive rate aur false positive rate ka tradeoff dikhati hai, balanced data mein theek. Heavily imbalanced data (fraud 0.1%) mein ROC-AUC dhoka de sakta hai kyunki true negatives bahut zyada hote hain. PR-AUC precision aur recall par focus karta hai, rare class ka sahi picture deta hai.

Analogy: Shehar mein 10 chor aur 10 lakh imaandar log. Police 1000 innocent pakad le to false positive rate bahut kam dikhti hai (0.1%) aur ROC achha lagta hai, lekin PR-AUC batata hai ki precision kharab hai.

33. SQL: RANK, DENSE_RANK aur ROW_NUMBER mein fark?

Detail: Tie hone par: **ROW_NUMBER** sabko unique number (1,2,3,4). **RANK** ties ko same rank, phir agla skip (1,1,3,4). **DENSE_RANK** ties same, skip nahi (1,1,2,3).

```
-- Second highest salary
SELECT salary FROM (
  SELECT salary,
    DENSE_RANK() OVER (ORDER BY salary DESC) AS rnk
  FROM employees
) t WHERE rnk = 2;
```

Analogy: Race mein do log ek saath pehle aaye. **RANK:** dono 1st, agla 3rd (Olympics jaisa). **DENSE_RANK:** dono 1st, agla 2nd. **ROW_NUMBER:** kisi ek ko zabardasti 1st, doosre ko 2nd.

34. WHERE vs HAVING, aur SQL joins?

Detail: WHERE grouping se pehle individual rows filter karta hai. HAVING grouping ke baad groups filter karta hai (aggregates par). Joins: INNER (sirf matching rows), LEFT (left ki saari rows plus matches), RIGHT (ulta), FULL (dono taraf ki saari).

```
SELECT dept, AVG(salary) FROM employees
WHERE active = 1
GROUP BY dept
HAVING AVG(salary) > 50000;
```

Analogy: WHERE = gate par entry check (kaun andar aayega). HAVING = andar team banane ke baad sirf wo teams rakho jinka average score 50 se upar hai.

35. Time series mein stationarity kyun chahiye?

Detail: Stationary series mein mean, variance aur autocorrelation time ke saath nahi badalte, isliye pattern predictable hota hai. ARIMA jaise models yehi assume karte hain. Stationary nahi hai to differencing ya log transform lagate hain, aur ADF test se check karte hain.

Analogy: Shant jheel mein lehren ka pattern samajhna aasaan hai. Agar paani level din-ba-din badhta ja raha ho (trend), to average lehr ki height ka koi matlab nahi. Pehle trend hatao, phir pattern samjho.

36. Multicollinearity kya hai?

Detail: Jab do ya zyada input features aapas mein highly correlated hote hain. Regression coefficients unstable ho jaate hain. Detect: correlation matrix, VIF (5 ya 10 se upar concerning). Fix: ek feature drop, features combine, PCA ya regularization (Ridge).

Analogy: Do log ek hi project par bilkul same kaam kar rahe hain. Aap nahi bata sakte ki success ka credit kisko jaaye.

37. App ka DAU 10% gir gaya, kaise investigate karoge?

Detail: 1. Pehle data/tracking bug check karo.

2. Segment karo: platform, geography, new vs old users, app version, channel.

3. Time dekho: kab se gira, suddenly ya slowly? Release, outage, festival, competitor launch?

4. Funnel dekho: kis step par users drop ho rahe hain?

5. Hypothesis banao, data se verify karo, recommendation do.

Analogy: Gaadi ka mileage achanak gir gaya. Mechanic pehle meter check karega (tracking bug), phir dekhega sirf highway par gira ya city mein bhi (segmentation), phir poochega kab se (recent change), tab engine, tyre, petrol check karega.

Interview mein bolo: Ye sawal dekhta hai ki aap structured soch sakte ho ya nahi. Soch ko zor se bolo.

38. Model ko non-technical stakeholder ko kaise explain karoge?

Detail: Math aur jargon nahi, business impact batao. Feature importance ya SHAP se dikhao kaun se factors prediction ko drive karte hain. Simple example do, aur limitations honestly batao.

Analogy: Doctor ye nahi kehta ki aapka logistic regression score 0.8 hai. Wo kehta hai aapka sugar aur BP dono high hain, isliye risk zyada hai, ye dawai aur parhez karo.

39. K-means mein K kaise chunte ho?

Detail: Elbow method, silhouette score, aur sabse zaroori: business interpretability. Dhyan rakho: K-means spherical clusters assume karta hai, scaling maangta hai, outliers se prabhavit hota hai.

Analogy: Kapde ki dukaan ke customers ko segments mein baantna hai. Math kahe K=9 best hai, lekin marketing team 9 campaigns nahi chala sakti. Practical K=3 ya 4.

40. Pehle model banate ho ya baseline? ML zaroori bhi hai?

Detail: Hamesha simple baseline pehle (rule-based, average ya logistic regression). Usse business goal poora ho jaye to complex ML ki zaroorat nahi. Complex model tabhi jab measurable gain ho, kyunki complexity ka cost hai: maintenance, explainability, latency.

Analogy: Do kilometre door dukaan jaana hai to cycle kaafi hai. Ferrari sirf isliye mat kharido ki cool hai.

Interview mein bolo: Mera pehla sawal hota hai: kya is problem ko ML ki zaroorat hai? Pehle simple baseline banata hoon aur use beat karne ka target rakhta hoon.

Part 3: Forward Deployed Engineer (20 Q)

41. FDE kya hota hai aur normal Software Engineer se kaise alag hai?

Detail: FDE customer ke saath milkar unki asli problem solve karta hai: requirements samajhta hai, product ko customer ke environment mein integrate/customize karta hai aur end-to-end deliver karta hai. SWE zyada tar product ka ek hissa banata hai aur customer se door rehta hai. FDE ka focus customer ka outcome hota hai. Coding, consulting aur problem solving ka mix.

Analogy: SWE = car factory ka engineer jo behtareen engine banata hai. FDE = mechanic-cum-advisor jo customer ke ghar aake gaadi unke raste aur garage ke hisaab se set karta hai.

Interview mein bolo: Mujhe code likhna aur customer ki problem solve hote dekhna dono pasand hain. Mere liye success ye hai ki customer ka kaam sach mein aasaan ho.

42. Customer ki problem vague ho to kaise approach karoge?

Detail: Pehle solution nahi, sawal: asli business problem kya hai? Abhi kaam kaise hota hai? Success kaise naapenge? Kaun stakeholders hain aur final decision kaun lega? Phir chhota scope tay karke jaldi prototype dikhata hoon, kyunki cheez dikhne par feedback milta hai aur clarity aati hai.

Analogy: Tailor ke paas koi bole mujhe achha sa kurta chahiye. Achha tailor turant kaat nahi deta: shaadi ya office? Kapda? Budget? Phir ek sample dikhata hai.

43. Customer ka data messy hai aur 2 din mein demo dena hai, kya karoge?

Detail: Poora data clean karna 2 din mein possible nahi aur zaroori bhi nahi.

1. Ek representative sample lo.
2. Sirf demo ke liye critical fields clean karo.
3. Thin end-to-end slice banao.
4. Demo mein clearly batao ki production ke liye kya baaki hai.

Analogy: Ghar mein mehmaan aane wale hain. Drawing room aur washroom chamkaoge, baaki ke baare mein honestly maante ho ki wo baad mein.

Interview mein bolo: Perfect demo nahi, working demo dunga, aur gaps ke baare mein transparent rahunga.

44. Customer ne aisa feature manga jo possible nahi ya bahut costly hai?

Detail: Seedha no nahi. Pehle underlying goal samjhata hoon (wo feature kyun chahiye?). Phir honest constraints batata hoon aur 2-3 alternatives deta hoon trade-offs ke saath (time, cost, accuracy), aur apni recommendation bhi.

Analogy: Customer bole 5 minute mein biryani chahiye. Waiter nahi ho sakta nahi bolta. Wo kehta hai biryani mein 40 minute lagenge, lekin pulao 15 minute mein mil sakta hai. Aap kya chahenge?

45. Customer ka data cloud mein nahi bhej sakte (security/compliance), kya karoge?

Detail: Options: on-prem ya customer ke VPC mein deployment; self-hosted ya open-source models; data masking/anonymization; minimal data access; audit logs aur access control. Security aur compliance team ko shuru se involve karna chahiye, end mein nahi.

Analogy: Bank ka locker. Bank sona bahar nahi le jaane deta, to kaam bank ke andar hi karte ho ya sirf zaroori cheez ki photocopy lete ho.

46. Third-party API integrate karte waqt kya dhyan rakhte ho?

Detail: - Authentication: OAuth ya API keys (keys code mein nahi, secrets manager mein).

- **Rate limits** aur throttling.

- **Retries with exponential backoff.**

- **Idempotency:** retry hone par duplicate action (double payment) na ho.

- Pagination, timeouts, error handling, logging.

Analogy: Bade office mein receptionist se kaam karwana. ID dikhana (auth), ek time par ek hi bande se milna (rate limit), busy ho to thodi der baad aana (backoff), ek hi form do baar jama na ho (idempotency).

47. Webhooks aur Polling mein fark?

Detail: Polling: hum baar-baar poochte hain kuch naya hai? (simple, lekin wasteful aur delay).

Webhook: event hone par server khud hamare endpoint ko notify karta hai (real-time, efficient).

Webhook endpoint reliable, secure (signature verify) aur idempotent hona chahiye, kyunki same event do baar aa sakta hai.

Analogy: Polling = har 5 minute mein delivery wale ko phone karke poochna mera parcel kahan hai.

Webhook = delivery wala khud darwaze par bell bajaye.

48. Customer site par production issue aaya, kaise debug karoge?

Detail: 1. Impact aur severity samjho, customer ko turant update do.

2. Quick mitigation (rollback ya workaround).

3. Logs, metrics aur recent changes se reproduce karo.

4. Root cause dhundh ke permanent fix.

5. Post-mortem likho.

Analogy: Ghar mein paani leak. Pehle main valve band karo (mitigation), phir pata lagao pipe kahan se phati (root cause), phir pipe badlo (fix), aur aage ke liye regular check lagao.

Interview mein bolo: Pehle bleeding rokta hoon, phir surgery karta hoon.

49. Prototype aur Production mein kya fark hai?

Detail: Prototype mein goal hai idea jaldi validate karna. Production mein chahiye reliability, security, monitoring, testing, error handling, scalability aur maintainability. Customer ko clearly batao ki demo ko production mat samjho.

Analogy: Ghar mein ek baar banayi maggi (prototype) aur restaurant mein roz 500 plates ek jaisi quality mein banana (production). Kaam bilkul alag.

50. LLM POC mein accuracy kam aa rahi hai, kya karoge?

Detail: Pehle failures ko categorize karo: retrieval galat, generation galat, ya data hi kharab?

- Chhota eval set banao.

- Chunking aur prompts improve karo.

- Hybrid search aur reranker add karo.

- Data quality sudharo.

- Zaroorat ho to better model ya fine-tuning.

Analogy: Student ke marks kam aaye. Teacher dekhta hai: kaun se sawal galat hue? Concept samajh nahi aaya, padha hi nahi, ya paper galat padha? Bina diagnosis ke aur padho kehna bekaar hai.

51. PII/sensitive data kaise handle karoge?

Detail: - Least-privilege access.

- Encryption in transit aur at rest.

- PII masking/redaction (naam, phone, Aadhaar) LLM ko bhejne se pehle.
- Logs mein sensitive data nahi.
- Retention policy.
- Compliance rules (GDPR, HIPAA, India mein DPDP Act).

Analogy: Hospital mein patient ki file har kisi ko nahi milti. Sirf us doctor ko jo ilaaj kar raha hai, locked cabinet se, entry register ke saath.

52. ETL pipeline kaise design karoge?

Detail: **Extract** (APIs, DB, files), **validate** (schema aur quality checks), **Transform** (clean, normalize), **Load** (warehouse). Dhyan rakho: idempotent jobs, error handling aur retries, monitoring aur alerts, incremental loads (roz poora data nahi, sirf naya).

Analogy: Doodh ka dairy plant. Kisanon se doodh aata hai (extract), quality test (validate), pasteurize aur packaging (transform), dukaanon tak (load). Kharab doodh pakda gaya to poora batch nahi, sirf wo hissa alag.

53. Docker kyun use karte hain aur deployment kaise karoge?

Detail: Docker app aur uski saari dependencies ko ek container mein package karta hai, to mere laptop par chal raha tha wali problem khatam. Flow: image build -> registry mein push -> Kubernetes ya cloud service par run. Secrets alag se, health checks aur rollback plan ke saath.

Analogy: Tiffin box. Khana, chammach, namak sab ek dabbe mein band. Kisi bhi office mein khol do, khana wahi rahega.

54. Customer ki expectations kaise manage karte ho?

Detail: Shuru mein clear scope aur timeline, regular updates (hafte mein ek baar), risks jaldi batana (surprise nahi), under-promise over-deliver, aur har important commitment likhit mein (email ya doc).

Analogy: Courier wala jo kehta hai kal tak pakka aur parso deta hai, usse log naraz hote hain. Jo 2-3 din kehta hai aur kal de deta hai, usse log pasand karte hain.

55. Customer gussa ho to kya karoge?

Detail: Pehle shaanti se suno, beech mein roke bina. Problem ko acknowledge karo (main samajh sakta hoon ye frustrating hai). Blame na do, na team ko, na customer ko. Phir concrete action plan aur timeline do, aur promise se pehle follow-up karo. Zyada tar gusse ka matlab hota hai ki customer ko lagta hai unki baat suni nahi ja rahi.

Analogy: Shikayat karne wale ko dukaandaar pehle chai pilaake baat sune, to aadha gussa wahin utar jaata hai.

56. Kai customers ek saath kaam maange to priority kaise tay karoge?

Detail: Har request ko 4 cheezon par dekhta hoon: impact, urgency, strategic value aur effort. Stakeholders se transparently baat karta hoon ki kya pehle hoga aur kyun. Jo kaam reusable hai (kai customers ko fayda) usse prefer karta hoon.

Analogy: Emergency ward mein doctor first-come-first-served nahi, severity ke hisaab se patients dekhta hai aur sabko batata hai ki kitna wait karna padega.

57. Customer feedback product team tak kaise pahunchate ho?

Detail: Sirf customer ko ye pasand nahi aaya nahi. Structured feedback: problem, kitne customers affected, business impact, current workaround aur suggested fix. Isse product team data ke saath

prioritize kar paati hai.

Analogy: Delivery partner sirf customers naraz hain nahi bolta. Wo bolta hai is area mein 30 orders mein 8 late hue, wajah road closure hai, aur ye solution ho sakta hai.

58. Project ki success kaise naapte ho?

Detail: Business metrics se: time saved, cost reduced, accuracy, user adoption, revenue impact. Kaam shuru hone se pehle baseline aur target tay karta hoon (report banane mein 4 ghante lagte hain, target 30 minute).

Analogy: Diet shuru karne se pehle weight naapna zaroori hai, warna 2 mahine baad pata hi nahi chalega ki fark pada ya nahi.

59. Ek aisi situation batao jahan requirements clear nahi thi (STAR format).

Detail: Ye sawal lagbhag pakka puchha jaata hai. STAR:

S (Situation): context kya tha?

T (Task): aapki zimmedari kya thi?

A (Action): aapne khud kya kiya? (main bolo, hum nahi)

R (Result): natija, numbers ke saath.

Template: Client ne bola hume AI chatbot chahiye lekin ye nahi bataya ki kis kaam ke liye (S). Mujhe use scope karke 2 hafte mein POC deliver karna tha (T). Maine support team ke saath baith ke dekha ki kis type ke tickets zyada aate hain, top 3 use cases choose kiye aur chhota prototype dikhaya (A). Result: [apne asli numbers] (R).

Analogy: Tailor ko bina naap ke kurta silna: pehle naap lo (clarify), sample banao (prototype), phir final silo.

Interview mein bolo: Agar real work example nahi hai to project ya academy ka example do. Jhooth mat bolo, follow-up sawal aate hain.

60. Aap FDE kyun banna chahte ho?

Detail: Generic jawab (mujhe challenges pasand hain) mat do. Apni asli wajah batao: coding aur customer interaction dono pasand hain, real-world messy problems solve karna achha lagta hai, jaldi seekhna aur ownership lena pasand hai. Ek personal example ke saath bolo.

Analogy: Achha trekking guide sirf rasta nahi jaanta, wo group ko samajhta hai, mausam dekhta hai aur manzil tak pahunchata hai. FDE bhi aisa hi guide hai.

Interview mein bolo: Mujhe building pasand hai, lekin sabse zyada khushi tab hoti hai jab mera banaya kaam kisi ke asli kaam mein aata hai. [Apna example].

Bonus: Interview ke din ke tips

- Pehle clarifying sawal poocho, seedha jawab mat do. Isse aap mature lagte ho.
- Soch ko zor se bolo. Interviewer reasoning dekhna chahta hai, sirf final answer nahi.
- Pata nahi ki jagah bolo: Mujhe directly nahi pata, lekin mera approach ye hoga...
- Resume ka har point ek sawal ban sakta hai. Projects ke decisions aur trade-offs taiyar rakho.
- Interview ke end mein 2 sawal poochho: Is role mein pehle 90 din mein success kaisi dikhegi? Team kaun se customers ke saath kaam karti hai?
- Formula yaad rakho: S-A-E (Short answer, Approach/reason, Example).